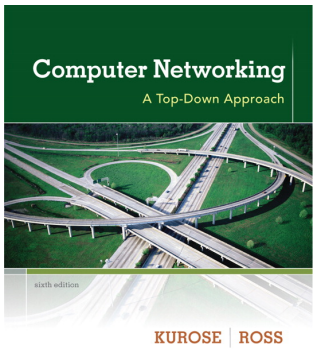
Wireshark Lab: DNS
SOLUTION

Supplement to *Computer Networking: A Top-Down Approach, 6th ed.*, J.F. Kurose and K.W. Ross



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1. Run *nslookup* to obtain the IP address of a Web server in Asia. What is its IP address? ***ANSWER:*** *I performed nslookup for www.rediff.com. Its IP address is 208.184.138.70*

```
C:\Documents and Settings>nslookup www.rediff.com
Server:  dns-prime.poly.edu
Address:  128.238.29.22

Name:    www.rediff.com
Address:  208.184.138.70
```

Screenshot taken for question 1

2. Run *nslookup* to determine the authoritative DNS servers for a university in Europe. What is its IP address. ***ANSWER****: I performed nslookup for a European University in Ioannina Greece. Its IP address is 128.238.29.22*

```
C:\Documents and Settings\andromahe>cd..

C:\Documents and Settings>nslookup -type=NS uoi.gr
Server:  dns-prime.poly.edu
Address:  128.238.29.22

Non-authoritative answer:
uoi.gr  nameserver = kouzina.noc.uoi.gr
uoi.gr  nameserver = marina.noc.uoi.gr
uoi.gr  nameserver = nic.grnet.gr

kouzina.noc.uoi.gr      internet address = 195.130.120.110
marina.noc.uoi.gr       internet address = 195.130.120.120
nic.grnet.gr    internet address = 194.177.210.210

C:\Documents and Settings>
```

Screenshot taken for question 2

3. Run *nslookup* so that one of the DNS servers obtained in Question 2 is queried for the mail servers for Yahoo! mail. What is its IP address? ***ANSWER:*** *the IP address of the mail server(s) is 18.72.0.3.*

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```
C:\Documents and Settings>nslookup mail.yahoo.com bitsy.mit.edu
Server:  BITSY.MIT.EDU
Address:  18.72.0.3

Non-authoritative answer:
Name:    login.yahoo.akadns.net
Address:  216.109.127.60
Aliases:  mail.yahoo.com, login.yahoo.com


C:\Documents and Settings>
```

Screenshot taken for question 3

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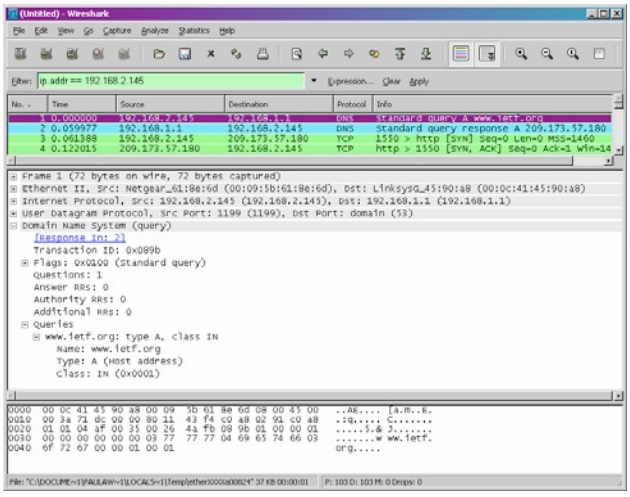
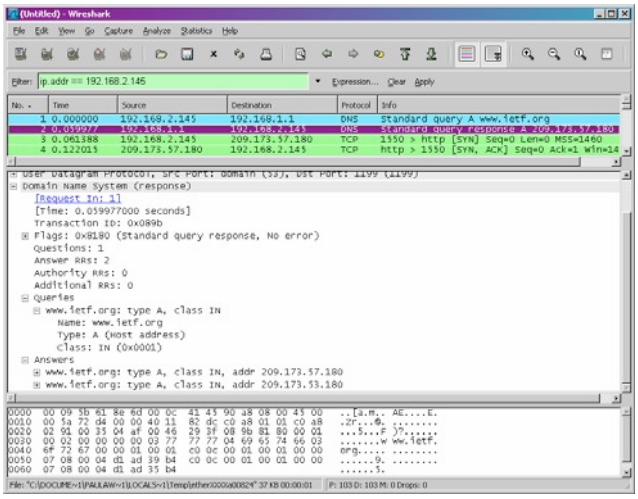
PART 3a



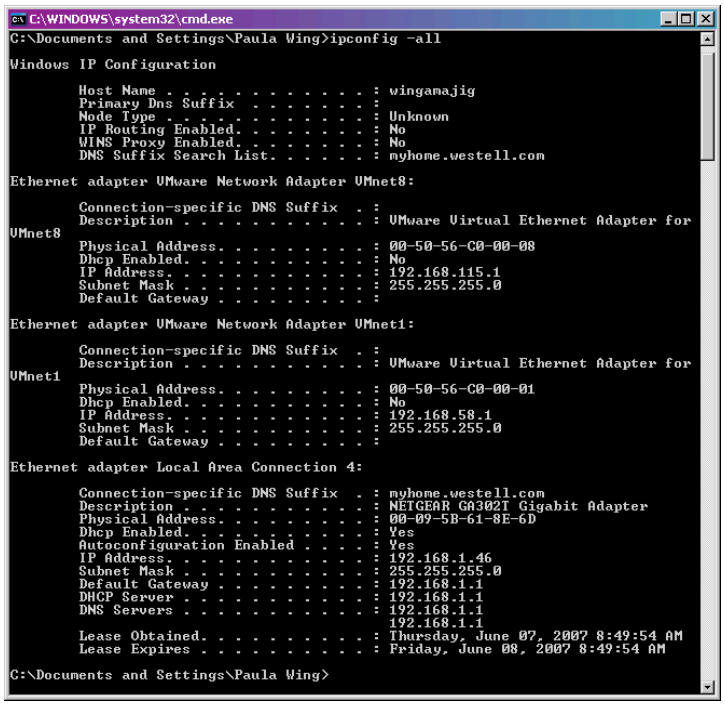
Screenshot for DNS query



Screenshot for DNS response

4. Locate the DNS query and response messages. Are then sent over UDP or TCP? ***ANSWER:*** *They are sent over UDP*

5. What is the destination port for the DNS query message? What is the source port of DNS response message? ***ANSWER:*** *The destination port for the DNS query is 53 and the source port of the DNS response is 53.*



```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\Paula Wing>ipconfig -all

Windows IP Configuration

        Host Name . . . . . . . . . . . . : wingamajig
        Primary Dns Suffix  . . . . . . . :
        Node Type . . . . . . . . . . . . : Unknown
        IP Routing Enabled. . . . . . . . : No
        WINS Proxy Enabled. . . . . . . . : No
        DNS Suffix Search List. . . . . . : myhome.westell.com

Ethernet adapter VMware Network Adapter VMnet8:

        Connection-specific DNS Suffix  . :
        Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for
VMnet8
        Physical Address. . . . . . . . . : 00-50-56-C0-00-08
        Dhcp Enabled. . . . . . . . . . . : No
        IP Address. . . . . . . . . . . . : 192.168.115.1
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet1:

        Connection-specific DNS Suffix  . :
        Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for
VMnet1
        Physical Address. . . . . . . . . : 00-50-56-C0-00-01
        Dhcp Enabled. . . . . . . . . . . : No
        IP Address. . . . . . . . . . . . : 192.168.58.1
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        Default Gateway . . . . . . . . . :

Ethernet adapter Local Area Connection 4:

        Connection-specific DNS Suffix  . : myhome.westell.com
        Description . . . . . . . . . . . : NETGEAR GA302T Gigabit Adapter
        Physical Address. . . . . . . . . : 00-09-5B-61-8E-6D
        Dhcp Enabled. . . . . . . . . . . : Yes
        Autoconfiguration Enabled . . . . : Yes
        IP Address. . . . . . . . . . . . : 192.168.1.46
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        Default Gateway . . . . . . . . . : 192.168.1.1
        DHCP Server . . . . . . . . . . . : 192.168.1.1
        DNS Servers . . . . . . . . . . . : 192.168.1.1
                                            192.168.1.1
        Lease Obtained. . . . . . . . . . : Thursday, June 07, 2007 8:49:54 AM
        Lease Expires . . . . . . . . . . : Friday, June 08, 2007 8:49:54 AM

C:\Documents and Settings\Paula Wing>
```

Screenshot for ipconfig –all

6. To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same? ***ANSWER:*** *It's sent to 192.168.1.1, which is the IP address of one of my local DNS servers.*

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7. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? ***ANSWER:*** *It's a type A Standard Query and it doesn't contain any answers.*

8. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain? ***ANSWER:*** *There were 2 answers containing information about the name of the host, the type of address, class, the TTL, the data length and the IP address.*

```
Answers
www.ietf.org: type A, class IN, addr 209.173.57.180
   Name:
   www.ietf.org
   Type: A (Host
   address) Class:
   IN (0x0001)
   Time to live: 30 minutes
   Data length: 4
   Addr: 209.173.57.180
www.ietf.org: type A, class IN, addr 209.173.53.180
   Name:
   www.ietf.org
   Type: A (Host
   address) Class:
   IN (0x0001)
   Time to live: 30 minutes
   Data length: 4
   Addr: 209.173.53.180
```

9. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? ***ANSWER:*** *The first SYN packet was sent to 209.173.57.180 which corresponds to the first IP address provided in the DNS response message.*

10. This web page contains images. Before retrieving each image, does your host issue new DNS queries? ***ANSWER:*** *No*

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